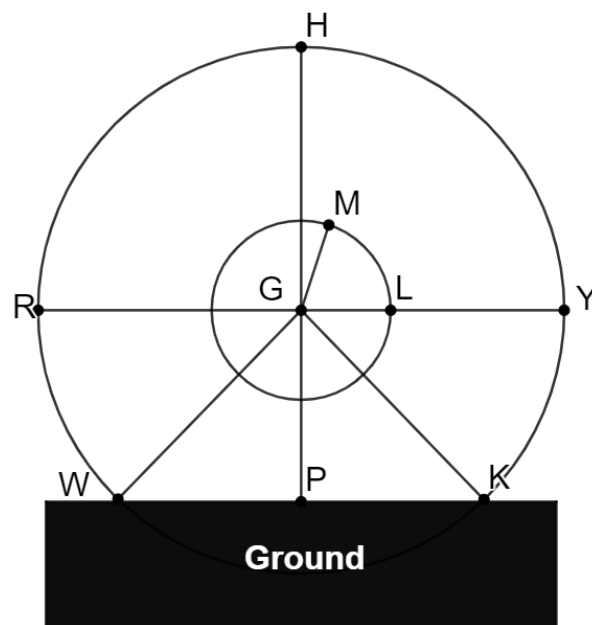


1. Guangzhou Circle is a circular building in Guangzhou, China designed by Italian architect Joseph di Pasquale. The building's height is 138 meters (m) and it has a diameter of 194 m. Guangzhou Circle has a circular hole at its center that is open with views of the Zhujiang River. The diameter of the circular hole is 48 m. A diagram of the building is shown where the smaller circle represents the circular hole, $\overline{GL}$ is the radius of the circular hole, $\overline{HP}$ is height of the building, and $\overline{RY}$ is diameter of the building.

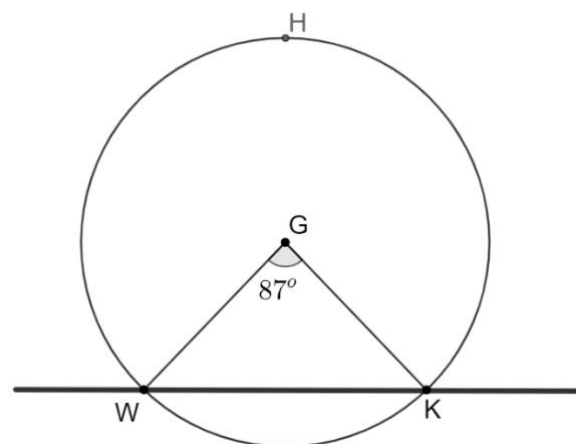


- a. Determine the area, in square meters, of the sector $\widehat{LM}$ if $m\angle LGM = 72^\circ$. Round your answer to the nearest thousandth.

area	
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- b. The part of the building that rests on the ground is not circular. In the diagram circle G represents the Guangzhou circle and $\overline{WK}$ represents the ground.

Determine the length of $\widehat{WK}$ if $m\angle W GK = 87^\circ$. Round your answer to the nearest thousandth meter.

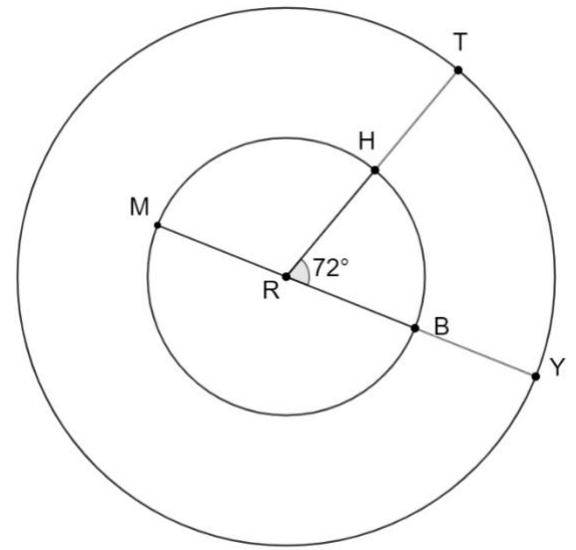


arc length	
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2. Two circles are shown where $RB = 27$ centimeters (cm), $HT = 25.5$ cm, $m\angle HRB = 72^\circ$.

- a. Determine the length of $\widehat{MH}$. Leave in terms of π .

arc length	
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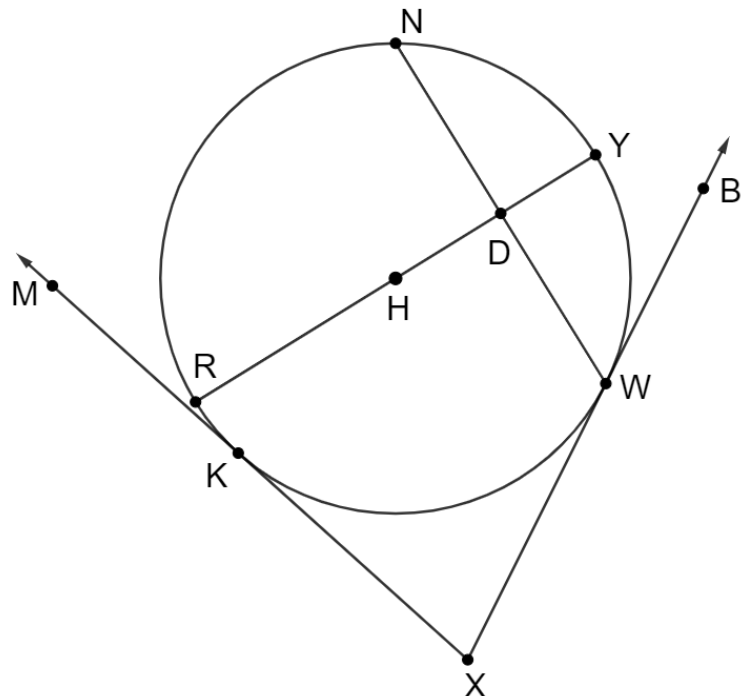


- b. Determine the length of $\widehat{TY}$. Leave in terms of π .

arc length	
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3. Circle H is shown where points N , Y , W and K are on the circle, $m\widehat{NY} = 58^\circ$, $m\widehat{RK} = 16^\circ$, $m\widehat{KW} = 106^\circ$, and $\overline{RY}$ is a diameter and a perpendicular bisector of $\overline{NW}$.

Determine $m\angle KXW$.



$m\angle KXW =$	
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6. Circle N is shown where $\triangle LNW$ is isosceles, $m\widehat{LH} = 105^\circ$, $m\angle LWN = 25^\circ$, and $m\angle DNW = 50^\circ$.

Determine each measure.

a. $m\angle LBH$

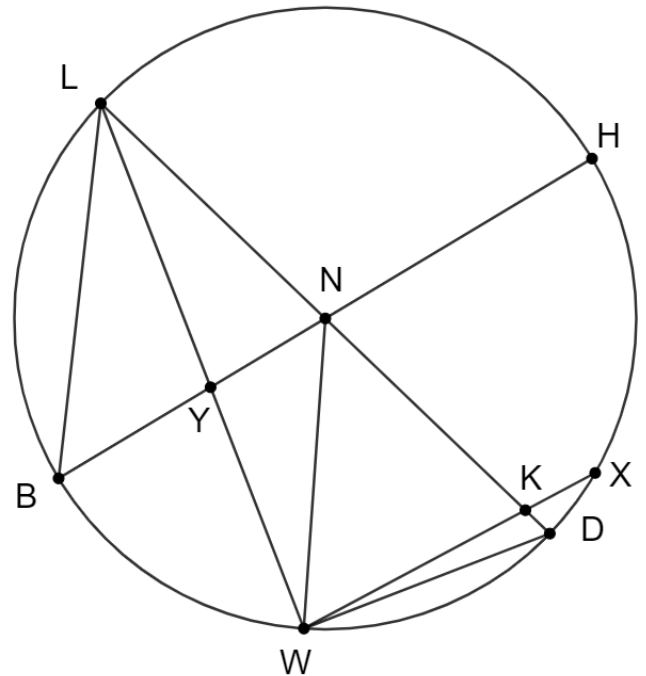
$m\angle LBH =$	
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b. $m\angle DWL$

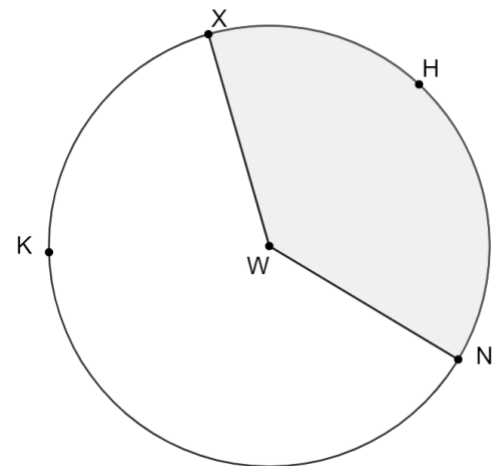
$m\angle DWL =$	
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c. $m\widehat{BW}$

$m\widehat{BW} =$	
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7. Circle W is shown where $m\widehat{XHN} = 137^\circ$ and $WN = 5.8$ centimeters. Determine the area of the shaded sector. Round your answer to the nearest thousandth.



area	
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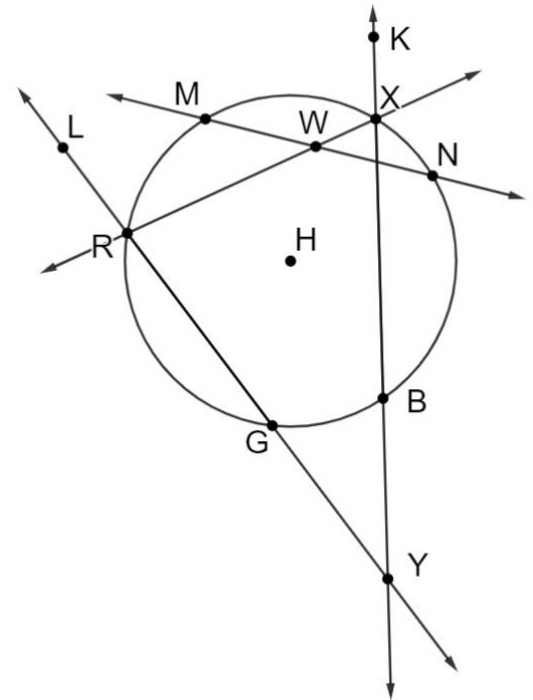
8. Circle H is shown where points $M, X, N, B, G,$ and R are on the circle and $m\widehat{RM} = (9x - 4)^\circ$, $m\widehat{MX} = (8x + 14)^\circ$, $m\widehat{NX} = (5x - 2)^\circ$, $m\widehat{GB} = 40^\circ$, and $m\angle GYB = (7x - 6)^\circ$.

- a. Determine the value of x .

$x =$	
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- b. Determine $m\angle XWN$.

$m\angle XWN =$	
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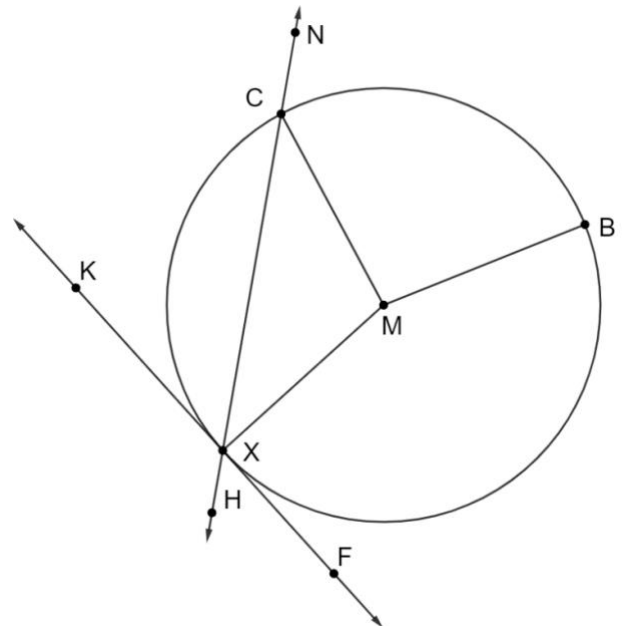
9. Circle M is shown where points $X, C,$ and B are on the circle and $m\angle XMC = (4y - 16)^\circ$, $m\angle BMC = (3y + 6)^\circ$, and $m\angle XMB = (5y + 10)^\circ$.

- a. Determine the value of y .

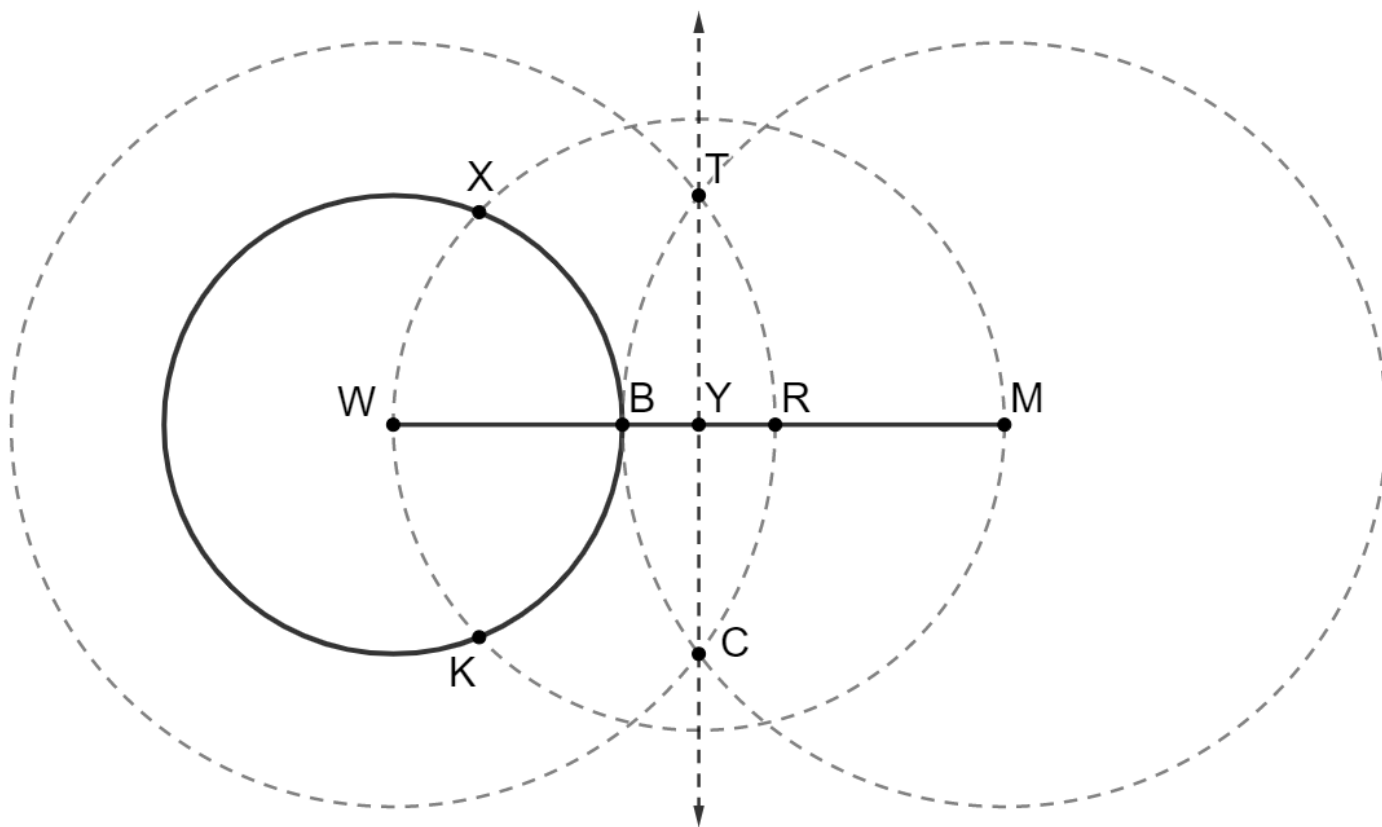
$y =$	
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- b. Determine $m\angle KXC$.

$m\angle KXC =$	
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10. An incomplete construction of a line tangent to circle W is shown. Points X , B , and K are on circle W . Points X , W , K , and M are on circle Y . Point Y is the midpoint of $\overline{WM}$ and $m\angle TYM = 90^\circ$.



Complete the statements.

Since
☐ $\overline{WB}$
☐ $\overline{WM}$
☐ $\overline{WR}$
☐ $\overline{WX}$
☐ $\overline{WY}$
 is a
☐ diameter
☐ radius
 of
☐ circle W ,
☐ circle Y ,

☐ $\angle WXB$
☐ $\angle WXM$
☐ $\angle WXY$
☐ $\angle YTX$
☐ $\angle YWK$
 is

inscribed in a semicircle making it a right angle. Therefore,
☐ $\overline{WB}$
☐ $\overline{WM}$
☐ $\overline{WR}$
☐ $\overline{WX}$
☐ $\overline{WY}$
 is perpendicular to

to
☐ $\overline{BX}$.
☐ $\overline{RX}$.
☐ $\overline{TK}$.
☐ $\overline{XM}$.
☐ $\overline{YK}$.
 This results in a radius that is perpendicular to a line passing through a

point on
☐ circle W ,
☐ circle Y ,
 making
☐ $\overline{BX}$
☐ $\overline{RX}$
☐ $\overline{TK}$
☐ $\overline{XM}$
☐ $\overline{YK}$
 tangent to
☐ circle W .
☐ circle Y .